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Influence of fiber volume ratio on hybrid effects of pultruded aramid/carbon hybrid fiber reinforced polymer composite bars: Static mechanical properties

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Abstract

To improve the mechanical performance of hybrid composites, this study fabricated aramid-carbon fiber-reinforced polymer (A/CHFRP) bars through the pultrusion process, utilizing aramid fibers as the outer layer and carbon fibers as the inner core. The effect of the volume ratio of aramid to carbon fibers on the hybrid effect of the static mechanical properties of the A/CHFRP was studied using tensile, bending, and shear tests. The study revealed that increasing the carbon fiber content enhanced the tensile strength and modulus of A/CHFRP; however, the material did not exhibit “pseudo-ductility.” The flexural stress-strain curve of A/CHFRP displayed a linear pattern akin to that of CFRP. While higher carbon fiber content improved the flexural strength and modulus, it resulted in a reduction in shear performance. This study presents a design method for A/CHFRP, and the resulting material can be used in bridge cables to offer improved impact resistance.

Highlights

- Aramid/carbon hybrid fiber-reinforced polymer composite bars were prepared by pultrusion.
- The “pseudo-ductility” of A/CHFRP is affected by interlaminar bonding properties.
- The tensile strength of A/CHFRP can be accurately predicted.
- The flexural properties of the A/CHFRP increased with increasing carbon fiber contents.
- The shear performance of A/CHFRP declined as the carbon fiber content increased.

KEYWORDS

aramid/carbon hybrid fiber-reinforced polymer composite bars, flexural properties, hybrid effect, pultrude, tensile properties

1 | INTRODUCTION

Cable-supported bridges are among the most common types of bridges used in civil engineering. They operate in

harsh environments and are subjected to complex forces.¹ Traditional bridge cables are typically created from high-strength steel. Cable-supported bridges face three major challenges in harsh environments: fatigue and corrosion,

weight limitations, and high carbon emissions. These challenges directly contribute to more frequent maintenance and replacement requirements, significantly driving up overall maintenance costs. First, fatigue and corrosion problems of steel cables are prominent under the coupling of forces and the environment.² Corrosion-fatigue damage occurs before the service life of a bridge, which may cause it to collapse.³ Additionally, the damage caused by corrosion is irreversible, necessitating the use of costly anti-corrosion materials or treatments, along with increased labor and equipment investments, which substantially escalate maintenance costs. Throughout a bridge's service life, the fatigue-induced corrosion of steel cables leads to significant maintenance and replacement expenses.⁴ Additionally, steel cables are heavy and prone to considerable sagging. When the span of a bridge is larger than 3000 m, its construction requirements cannot be satisfied.⁵ The structural sagging caused by the weight of steel cables may necessitate additional reinforcement measures or design modifications. Furthermore, to meet the specific demands of long-span structures, the use of steel cables often requires more advanced construction techniques and specialized equipment, further driving up both construction and long-term maintenance costs. Third, carbon emissions from the production of traditional steel and cement building materials reach 2 billion tons per year, accounting for more than 20% of the total carbon emissions,⁶ posing an urgent demand for long-service life construction materials. Steel and cement production, transportation, and construction contribute significantly to carbon emissions.⁷ During the usage phase, bridge maintenance activities, such as anti-corrosion treatment for steel structures and concrete repairs, often require substantial material replenishment, further increasing the carbon footprint. Additionally, in the demolition phase, the disposal of waste materials (e.g., recycling or landfilling of steel and concrete) also contributes to additional carbon emissions. Since the environmental life of steel cable materials is limited (prone to corrosion, fatigue, etc.), they need to be replaced repeatedly during the use of the bridge. This not only increases direct maintenance costs but also has a further negative impact on the ecological environment due to high-frequency replacement, leading to increased indirect costs. In order to achieve low-carbon and environmental protection goals, improve material performance and extend service life, reduce long-term costs, and meet the needs of new bridge design, current bridge steel cables face many bottleneck problems that must be solved urgently.

In the 1980s, Urs Meier proposed the idea of constructing an 8400-meter-span CFRP cable bridge over the Strait of Gibraltar, which initiated research into the

application of CFRP cables for long-span bridge projects.⁸ Compared to traditional steel cables, CFRP cables offer significant advantages in terms of fatigue resistance, maintenance, and structural weight. CFRP cables exhibit superior fatigue resistance,⁹ as they are less prone to damage under cyclic loading and environmental factors, ensuring a longer service life. Their excellent corrosion resistance minimizes the need for frequent maintenance, significantly reducing lifecycle costs.¹⁰ Additionally, CFRP cables not only reduce the overall structural weight but also mitigate sagging effects, making them particularly advantageous for ultra-long-span bridges.¹ These characteristics make CFRP cables a highly efficient and sustainable choice for modern bridge construction. Substituting conventional steel cables with CFRP cables addresses the corrosion-fatigue issues of steel cables. This also allows for longer bridge cable spans, lowers maintenance costs, and reduces the need for frequent cable replacements.^{11,12} Currently, key challenges such as anchorage, static performance, vibration characteristics, damping performance, seismic performance, and wind resistance of CFRP bridge cables have been solved.¹³

Bridge cables also encounter the threat of low-speed impacts (vehicle collisions, falling rocks, etc.) during their production and service. Statistics on the reasons for replacing bridge cables show that vehicle collisions are the third largest cause of damage to steel cables, second only to fatigue, corrosion, fracture, and excessive cable forces.¹⁴ Low-speed impacts can cause internal damage to composite materials, such as debonding of composite interfaces, brittle fracture of resins, and even fiber breakage, resulting in a decrease in structural stiffness, stability, and bearing capacity, and are considered some of the most severe threats to composite components and structures.¹⁵ The transverse shear strength of a CFRP cable is only 10% of its longitudinal tensile strength. CFRP has a small elongation at break and poor fracture toughness, resulting in its limited deformation capacity.¹⁵ Besides, since the transverse shear strength of carbon fiber composites is significantly lower than their longitudinal tensile strength, they are prone to local damage under transverse forces or impact conditions, such as matrix cracking, fiber-matrix interface debonding, and fiber breakage. These localized damages weaken the overall mechanical properties of the material and may further develop into macroscopic failures. The low transverse shear strength also results in poor anchoring performance, making shear failure more likely to occur at the anchor exit.¹⁶ In addition, existing studies have shown that applying pre-tension to CFRP cables reduces their impact resistance. Therefore, the poor impact resistance of CFRP cables is an inevitable and urgent problem in bridge applications.^{17,18}

Fiber hybridization is a proven approach to enhance the performance of CFRPs.¹⁹ Hybridizing two different kinds of fibers can balance the performance of the two fibers.²⁰ This is an efficient method for improving and designing CFRP performance.²¹ Glass fiber reinforced polymer composites (GFRP) have excellent impact resistance.^{22,23} Existing research shows that mixing carbon fibers with aramid or glass fibers with better toughness can effectively improve the impact properties of CFRPs.²⁴ However, the low fatigue resistance of GFRP indicates that glass/carbon fiber hybrid composites fail to meet the performance requirements for bridge cables.²⁵ In addition, glass fiber has poor corrosion resistance and is no longer suitable for use in cross-sea bridges.⁵ Aramid has excellent impact resistance and is widely used for bulletproofing. Additionally, aramid has high tensile strength and modulus (~ 2.8 GPa, ~ 109 GPa), stable chemical properties, and excellent corrosion-fatigue resistance.²⁶ Mixing CFRPs with aramid fibers is expected to improve their impact performance and satisfy the impact resistance requirements of bridge cables.²⁷ Aramid fibers contribute to absorbing impact energy and reducing crack propagation.²⁸ Previous studies indicate that bidirectional A/CHFRP obtained a great deformation capacity.²⁹

Hybrid fibers are employed to harness the combined effects of two fibers, resulting in a performance that surpasses that of polymer composites reinforced with a single fiber.^{30,31} The hybrid effect is commonly defined as the increased apparent failure strain of hybrid composite compared to that in a non-hybrid fiber-reinforced composite.³² A positive hybrid effect is defined as a certain property of hybrid fiber-reinforced polymer composites (HFRPs) (such as elongation at break and strength) that is higher than the blending (reinforcement) properties of the two fiber reinforcements, and vice versa is a negative hybrid effect.³³ Therefore, designing a hybrid method can fully utilize the material properties of the two fibers to obtain a hybrid composite material with excellent properties of the two fibers.³⁴ The static mechanical properties of A/CHFRP bars are influenced by the fiber mixing methods and volume ratios, leading to variations in the mechanical performance of the hybrid composite.³⁵ Therefore, the hybrid effect on the static mechanical properties of A/CHFRP bars should be studied, and an optimal design method for A/CHFRP bars with excellent mechanical properties and a positive hybrid effect should be proposed. Research by Zachariah et al. showed that, compared to CFRPs, incorporating aramid fibers, which have lower density and higher toughness, reduces the modulus without significantly

affecting the tensile strength or specific strength. Additionally, the ultimate failure strain increases by 60%–75% relative to that of CFRPs.³⁶ The failure strain of A/CHFRPs whose carbon fiber volume fraction is 22.7% is 54.09% higher than that of CFRP materials.³⁷ Song³⁸ observed that the tensile strength and stiffness of hybrid composites with aramid fibers on the surface are superior to those with carbon fibers on the surface. Regarding flexural performance, Bao et al.³⁹ observed that increasing the carbon fiber content can improve the bending properties of A/CHFRP plates.

However, all previous studies studied A/CHFRP laminates, mostly focusing on bidirectional A/CHFRP, whereas FRP bars for bridge cables are frequently pultruded unidirectional composites. The molding method, fiber direction, and interface shape are different from those of laminates, and the hybrid effect of their static mechanical properties is unclear. The hybrid effect of the static mechanical properties of A/CHFRP bars must be studied. This study produced hybrid composite bars with a skin-core structure, incorporating aramid fibers in the outer layer and carbon fibers in the core. The static mechanical performance of the A/CHFRP bars was affected by the volume proportions of aramid and carbon fibers. An investigation was carried out to examine how the aramid-to-carbon fiber volume ratio influenced the hybrid effect on the composite's static mechanical properties. The results provide a theoretical basis for designing impact-resistant A/CHFRP bars for cable applications.

2 | MATERIALS AND METHODS

2.1 | Materials

The carbon fiber was T700 (SYT49S), produced by Zhongfu Shenying Carbon Fiber Co., Ltd. (Jiangsu Province, China). The density of carbon fiber is 1.8 g/cm^3 , and carbon fiber can be used stably at less than 600°C in air. The aramid was a para-aramid produced by Yantai Taihe New Materials Co., Ltd. (Shandong Province, China), with a tensile strength of 19.39 cN/dtex , a modulus of 108 GPa , and an oiling rate of 2.71% . The density of aramid is 1.44 g/cm^3 , and aramid can be used stably at less than 300°C in air. The resin used was epoxy resin produced by Hertz New Materials (Jiangsu Province, China), with a glass transition temperature of 235°C .

2.2 | Sample preparation

A/CHFRP bars with different fiber volume ratios and diameters of 5 mm were prepared via pultrusion. In the

production process of aramid fiber, the surface of aramid fiber contains a large amount of oil, which is not conducive to the combination with the resin matrix and is an isolation barrier. However, the oil will decompose at high temperatures, so before preparing aramid, a high-temperature furnace is used to remove the oil covering the surface of the aramid fiber to increase the bonding properties. A high-temperature furnace was added to the pultrusion process line to partially volatilize the oil agent before the aramid was impregnated with the resin. The temperature of the high-temperature box was 180°C. The pultruded samples used in this study were purchased from Zhongfu Carbon Fiber Core Cable Technology Co., Ltd. (Jiangsu Province, China). The production process diagram can be found in Mackin C D's research.⁴⁰

In hybrid composites, the tough fiber layer is typically placed on the exterior to efficiently absorb impact energy, while the carbon fiber layer is located internally to provide structural reinforcement.⁴¹ Therefore, the hybrid composites with skin-core structures can obtain better impact performance. In this study, CFRP bars, AFRP bars, and five different types of A/CHFRP bars with skin-core structures were fabricated, where aramid fiber formed the skin layer and carbon fiber served as the core. As shown in Table 1, the volume fractions of the aramid fibers were 0%, 10%, 30%, 50%, 70%, and 100%, and they were named CFRP, HFRP-10, HFRP-30, HFRP-50, HFRP-70, and AFRP, respectively. The fiber volume fraction of the seven types of composite bars prepared in this study was 73%.

2.3 | Tensile test

The tensile properties of CFRP, AFRP, and A/CHFRP were tested according to the ADTM D7205 code. The tensile tests of the composites were carried out in an electronic universal material testing machine (WDW-100 E; Jinan Times Testing Machine Co., Ltd., Jinan, China) at room temperature. The specimen was tested using a

support anchor, which has been proven to have excellent anchoring effects in previous studies, and no slippage occurred during the test. The diameter of the CFRP, AFRP, and A/CHFRP bars is 5 mm. The specimen length, anchor length, and test speed were 680 mm, 150 mm, and 5 mm/min, respectively. Repeat the test for 5 samples in each group. By evaluating the tensile strength and elastic modulus of CFRP, AFRP, and A/CHFRP, the synergistic interaction between aramid and carbon fibers in hybrid composites becomes evident, highlighting their enhanced axial load-bearing capacity. The tensile strength was calculated by Equation (1):

$$\sigma = \frac{F}{A} \quad (1)$$

where F is the ultimate tensile breaking force and A is the area of the specimen's cross section.

2.4 | Flexural test

The flexural properties of the CFRP, AFRP, and A/CHFRP were tested according to ASTM D4476. The composite bars were cut according to the test specifications and subjected to bending tests. The thickness of the cut specimen was less than the radius of the FRP bars and was kept constant. The flexural tests of the composites were carried out in an electronic universal material testing machine (WDW-100 E; Jinan Times Testing Machine Co., Ltd., Jinan, China) at room temperature. In the three-point bending test, the two support points were on a curved surface, the force was loaded onto the cut plane, and the test rate was 2 mm/min. The length of the samples is 300 mm and the span is 60 mm, with 5 specimens in each group. Flexural tests provide insights into the strength and stiffness of the composite under localized bending loads. The load-sharing behavior of hybrid fibers during bending deformation can be assessed by analyzing the bending modulus and failure modes.

TABLE 1 Basic parameters of aramid/carbon fiber hybrid-reinforced polymer composite bars.

Type	Volume ratio of aramid to all fibers (%)	The volume ratio of carbon fiber to total fiber (%)	Volume fraction of aramid (%)	Volume fraction of carbon fiber (%)
CFRP	0	100	0	73
HFRP-10	10	90	6.64	66.36
HFRP-30	30	70	16.85	56.15
HFRP-50	50	50	24.33	48.67
HFRP-70	70	30	30.06	42.94
AFRP	100	0	73	0

2.5 | Shear test

The shear properties of the CFRP, AFRP, and A/CHFRP were tested according to GB/T 30022-2013. The shear tests of the composites were carried out in an electronic universal material testing machine (WDW-100 E; Jinan Times Testing Machine Co., Ltd., Jinan, China) at room temperature. The shear test in this study was performed perpendicular to the fiber direction. The sample was fixed on a shear fixture according to the test specifications, and the upper blade was loaded downward perpendicular to the FRP. The diameters of the upper and lower blades matched those of the FRP. The length of the specimen is 300 mm, with 5 specimens in each group.

$$\tau = \frac{F}{2A} \quad (2)$$

where F is the ultimate tensile breaking force and A is the area of the specimen cross section.

3 | RESULTS AND DISCUSSION

3.1 | Tensile properties

Figure 1 presents the stress–strain curves for CFRP, AFRP, and A/CHFRP. The CFRP exhibited a linear elastic stress–strain curve, while the AFRP demonstrated an almost linear elastic behavior. The curve of the A/CHFRP was between those of the CFRP and AFRP and also exhibited linear elasticity. The strength and elastic modulus of the A/CHFRP declined, while the ultimate failure strain showed an upward trend with the carbon content

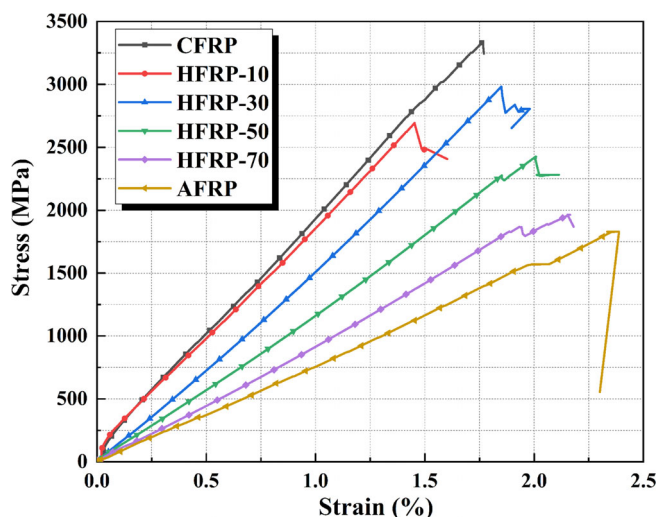


FIGURE 1 Tensile stress–strain curve of fiber-reinforced polymer composites.

from 10% to 70%. The ultimate failure strain of HFRP-10, HFRP-30, HFRP-50, and HFRP-70 was increased -17.60% , 8.39% , 13.83% , and 21.96% compared with CFRP, respectively. According to the definition of the hybrid effect of the ultimate failure strain by Swolfs et al.,³⁴ the hybrid effect is beneficial when the strain exceeds the ultimate failure strain of the CFRP. The hybrid effect of the ultimate failure strain can be calculated using Equation (3):

$$HE\% = 100 \times (\varepsilon_H - \varepsilon_C) / \varepsilon_C \quad (3)$$

where ε_H is the ultimate failure strain of the A/CHFRP, and ε_C is the ultimate failure strain of the CFRP.

Specifically, when the ultimate failure strain of the A/CHFRP exceeds that of CFRP, it is classified as a positive hybrid effect. According to calculations, the failure strain confounding effects of HFRP-10, HFRP-30, HFRP-50, and HFRP-70 were -17.60% , 8.39% , 13.83% , and 21.96% , respectively. This was consistent with the typical range of tensile failure strain hybrid effects proposed by researchers, which ranges from 10% to 50%.⁴¹ When the content of aramid fiber is 10%, the failure strain of A/CHFRP demonstrated a negative hybrid effect. After that, the failure strain gradually improved. Analysis indicates that when the aramid-to-carbon fiber volume ratio is below 30%, A/CHFRP exhibits a negative hybrid effect, whereas ratios exceeding 30% lead to a positive hybrid effect.

Generally, the strength of fibers in FRPs conforms to the Weibull distribution. The Weibull distribution is a commonly used statistical tool for describing the strength distribution of fibers in fiber-reinforced composites (FRPs). In carbon fiber composites, some weaker carbon fibers may fail prematurely by reaching their ultimate failure strain due to insufficient strength. Under sustained tensile loading, these weaker carbon fibers progressively fracture. When the number of fractured fibers accumulates to form a critical fracture cluster, the composite material as a whole will fail.⁴² During the stretching process, the fibers with large defects and low strength are damaged first. When the number of failed fibers reaches a sufficient level, the FRP fails.⁴³ In FRP, the tensile strength of fibers is heavily influenced by the size and distribution of internal defects, such as microcracks and voids, as these defects serve as primary initiation sites for fiber breakage. Larger defects result in reduced effective load-bearing areas, leading to lower tensile strength. Such defects are more likely to trigger fractures under lower stress, whereas smaller defects typically cause failure only under higher stress levels.⁴⁴

During the tensile process of unidirectional hybrid fiber-reinforced composites (HFRP), the variability in

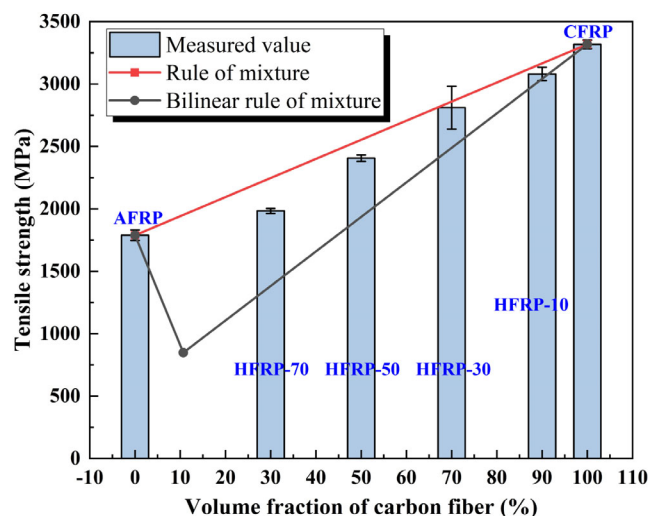
fiber strength causes weaker fibers to fracture first at lower strains. As the accumulation of fiber fractures reaches a critical number and forms a critical fracture cluster, the load can no longer be effectively redistributed among the remaining intact fibers, ultimately leading to the overall failure of the composite. The formation of a critical fracture cluster signifies that the cumulative effect of localized fiber fractures has reached the composite's load-bearing limit, acting as the trigger point for failure. For unidirectional HFRPs, under the action of a tensile load, the two types of fibers work together in the early stage, and subsequently the fibers with lower failure strain begin to break and fail. When the number of broken fibers reaches a certain level, a critical fracture cluster is formed, which triggers the failure of the FRP layer with a lower failure strain.⁴⁵ This can be divided into the following scenarios: (1) Energy is released at the moment of failure of the low-strain fiber layer. This energy is released between the low- and high-strain fiber layers. When the content of low-strain fibers is high or the interlayer bonding performance is low, interlayer failure occurs. After interlayer failure, stress cannot be transferred, resulting in only high-strain fibers bearing loads. The high-strain layer gradually forms a fracture cluster under the action of the load, which in turn causes the failure of the high-strain layer. (2) The low-strain fiber layer fails to cause interlayer debonding at the moment of failure. Under continuous tensile loading, the low-strain fiber layer, after fracturing, can still bear stress up to its limit at a certain distance from the fracture point. This behavior results in the appearance of "pseudo-ductility" on the stress-strain curve. Eventually, the stress reaches the failure strain of the high-strain fiber layer, leading to the formation of a critical fracture cluster within the high-strain layer, which ultimately causes failure.^{46,47}

In hybrid composite materials displaying a negative hybrid effect in failure strain, a high proportion of carbon fibers results in the formation of critical fracture clusters within the weaker carbon fibers, leading to the premature failure of the composite. This results in a negative hybrid effect. However, As the carbon fiber content decreases, the ultimate failure strain progressively increases. After the carbon fibers fail, the aramid fibers take over the load-bearing role and continue to sustain the stress until they ultimately reach their failure point. At this point, the failure strain demonstrates a positive hybrid effect. As the carbon and aramid fibers broke, the critical fracture cluster size in the FRP was quickly reached, which caused the HFRP to fail.⁴⁸ When the carbon fiber content was high, the failure strain of A/CHFRP demonstrated a negative hybrid effect. Due to the premature failure induced by the formation of a

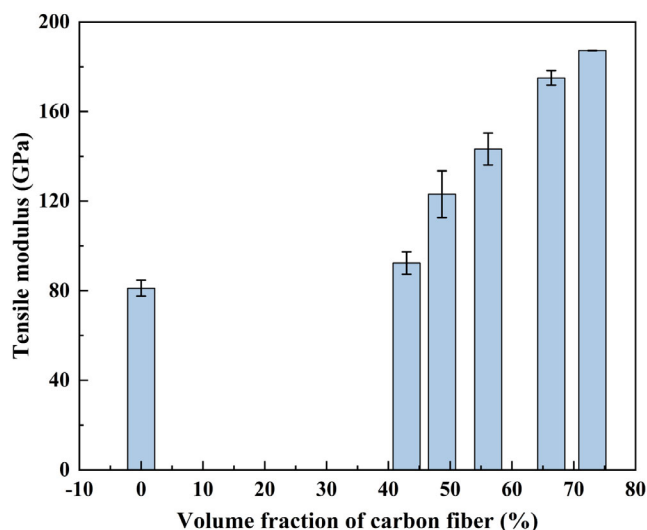
critical fracture cluster, the full performance potential of certain carbon fibers was not realized. As a result, the strength of HFRP-10 was found to be lower than that of HFRP-30, despite the latter having a lower carbon fiber content.

Additionally, When the proportion of carbon fiber occupies a larger volume, that is, when the aramid content is low, the energy released after the carbon fiber layer broke not only caused debonding between the layers but also caused some aramid fibers to break. During the continuous loading process, the aramid fibers gradually formed fracture clusters. Therefore, when the carbon fiber content was higher, more aramid fibers were damaged after fracture, and the stress transferred to the aramid fiber layer was higher. This results in a faster formation of fracture clusters in the aramid fiber layer, resulting in a lower failure strain for A/CHFRP than for CFRP. In this case, the A/CHFRP exhibits a negative hybrid effect. The failure strains of HFRP-30, HFRP-50, and HFRP-70 showed a positive hybrid effect. When the A/CHFRP reached the ultimate failure strain of the CFRP under tensile loading, the carbon fiber layer failed. Higher carbon fiber content released more energy upon failure, transferring greater stress to the aramid fiber layer, which increased the likelihood of its failure. Consequently, as carbon fiber content decreases, the ultimate failure strain of HFRP improves. The strength of the HFRP is influenced by both the carbon fiber and the aramid fiber, as shown in Equation (2). Based on Equation (2) and the subsequent analysis, an increase in aramid fiber content leads to a decrease in the overall strength of the HFRP. For HFRP-70, although the carbon fiber content was lower than that of aramid fiber, the modulus of carbon fiber and aramid fiber was significantly different. Therefore, for the A/CHFRP, which requires greater deformation, the carbon fiber content must be reduced to ensure the stress transfer of the A/CHFRP. When the carbon fiber content decreases, the energy released from the failure of the carbon fiber layer does not cause interlayer debonding. Under further tensile loading, the internal stress within the HFRP is redistributed. At locations far from the carbon fiber failure points, all fiber layers continue to share the load, progressively failing until the HFRP reaches its ultimate failure. At this stage, the final strength of the HFRP is controlled by the combined contributions of both the aramid fiber and the carbon fiber.

Figure 2 shows the strength and modulus of the A/CHFRP. The law of mixtures is typically used to calculate the Young's moduli of composites. In the early stage of tensile loading, two different fibers can work together; thus, the modulus can be calculated by a simple law of mixtures. For hybrid composites, the upper limit of their



(A) Tensile strength



(B) Tensile modulus

FIGURE 2 (A) Tensile strength and (B) Tensile modulus of fiber-reinforced polymer composites.

strength occurs when the two fibers always work together, and neither the layers nor the fibers stops working. Therefore, the law of mixtures can be used to express the upper limit of the strength of the hybrid composites, which can be expressed by Equation (4):

$$\sigma_H = \sigma_C V_C + \sigma_A V_A \quad (4)$$

where σ_C is the strength of the CFRP, V_C is the volume fraction of the CFRP, σ_A is the strength of the AFRP, and V_A is the volume fraction of the AFRP.

According to the introduction above, for the A/CHFRP, under a tensile load, the CFRP breaks first. As a result, two possible scenarios may arise. In the first scenario, when the content of carbon fiber is reduced, the carbon fiber ceases to function after fracturing but does not cause damage to the aramid fiber. The aramid fiber continues to bear the load and finally reaches the ultimate failure strain of the aramid fiber and fails. Therefore, the controlling factor for the final strength is the AFRP.⁴¹ This can be expressed as (5).

$$\sigma_H = \sigma_A V_A \quad (5)$$

In the second scenario, when the volume fraction of the CFRP is high, the carbon fibers fracture and cause the simultaneous failure of the AFRP. Here, the tensile strength of the A/CHFRP will be controlled by the failure strain of the carbon fiber. The strength of the hybrid composite is equal to the stress borne by the hybrid composite when the failure strain of the CFRP is reached. This can be expressed as Equation (6).

$$\sigma_H = \sigma_C V_C + \varepsilon_C E_A V_A \quad (6)$$

The critical volume fraction of the carbon fibers can be calculated using Equations (5) and (6): This bilinear model simplifies the tensile process of the A/CHFRP. The interlayer bonding performance of the carbon and aramid fiber layers is not considered. The bilinear model does not account for stress redistribution within the hybrid composite material after the carbon fiber layer fails. According to the bilinear model, once the carbon fiber layer is damaged, the interlayer is assumed to be completely destroyed, resulting in the carbon fiber layer fully withdrawing from load-bearing and no longer contributing to the composite's strength. Another scenario described by the model is that, following the failure of the carbon fiber layer, the aramid fiber layer is also immediately damaged and ceases to bear any load. Both scenarios represent idealized conditions and the worst-case failure modes, making them suitable as the lower limit for strength prediction. Therefore, this model can be used as the lower limit for predicting the tensile strength of the A/CHFRP.

As shown in Figure 2A, the strength of the A/CHFRP was within the triangular area formed by the mixing law and bilinear model. As the carbon fiber content increased, the strength of A/CHFRP improved. Compared to AFRP, the tensile strength of HFRP-10, HFRP-30, HFRP-50, and HFRP-70 increased by 11%, 27.78%, 52.78%, and 66.67%, respectively. Figure 2B illustrates that the modulus of A/CHFRP increases as the volume fraction of carbon fiber rises. Based on the rule of mixtures, when the fiber-to-resin volume ratio in the hybrid

composite remains unchanged, the overall modulus of the hybrid composite is determined by the combined contributions of the aramid layer, carbon fiber layer, and resin. Given the higher modulus of carbon fiber, an increased proportion of carbon fiber within the total fiber content results in a corresponding increase in the composite's modulus. The tensile properties of A/CHFRP are better than those of glass/carbon fiber hybrid reinforced polymer (G/CHFRP) composites. Rui Guo et al.²⁵ prepared the G/CHFRP, for which the volume fraction ratio of carbon to glass fiber was 2:3, and the tensile strength and tensile modulus of G/CHFRP were 975.5 MPa and 59GPa.

The failure modes of the A/CHFRP are shown in Figure 3. The CFRP displayed a typical explosive failure accompanied by a distinct brittle fracture behavior. For A/CHFRP, failure was characterized by internal carbon fiber rupture and damage to both the aramid and carbon fiber composite layers. As shown in Figure 3, the HFRP demonstrated noticeable interlayer debonding between the surface aramid fiber layer and the inner carbon fiber layer. After failure, the surface layer and the core layer peel off, revealing the morphology of the internal carbon fibers. This further confirms that a high carbon fiber content in the inner layer leads to interlayer debonding between the aramid and carbon fiber layers. Because HFRP-10 and HFRP-30 had high carbon fiber and low aramid contents, apparent interlaminar failure and aramid fiber fracturing caused by carbon fiber damage were observed from

the failure morphology. The explosive fracture presented in the failure morphology of HFRP-50 and HFRP-70 was not as noticeable as that of HFRP-10 and HFRP-30, but some interlaminar failure occurred between the AFRP and CFRP, and the failure mode did not exhibit the same brittle fracturing as the CFRP. The failure mode of the AFRP was completely different from that of the CFRP. AFRP did not exhibit a fiber explosion mode.

3.2 | Flexural properties

Figure 4 shows the flexural properties of the A/CHFRP. Figure 4A presents the flexural stress–strain curves for CFRP, AFRP, and A/CHFRP. The AFRP exhibited a non-linear flexural stress–strain curve with significant deformation capacity, while the curves for CFRP, HFRP-10, HFRP-50, and HFRP-70 were nearly linear. As the carbon fiber content increased, the failure strain of A/CHFRP showed a gradual rise, although it remained lower than that of CFRP. The notable modulus disparity between the carbon fiber and aramid fiber layers negatively affects the bending performance of A/CHFRP, weakening the interaction between the fiber layers and leading to a considerable reduction in bending strength. Under bending loads, the bending failure of A/CHFRP is primarily governed by the compressive properties of the surface carbon fiber layer. Figure 7B shows the flexural strength and modulus of the A/CHFRP. With the

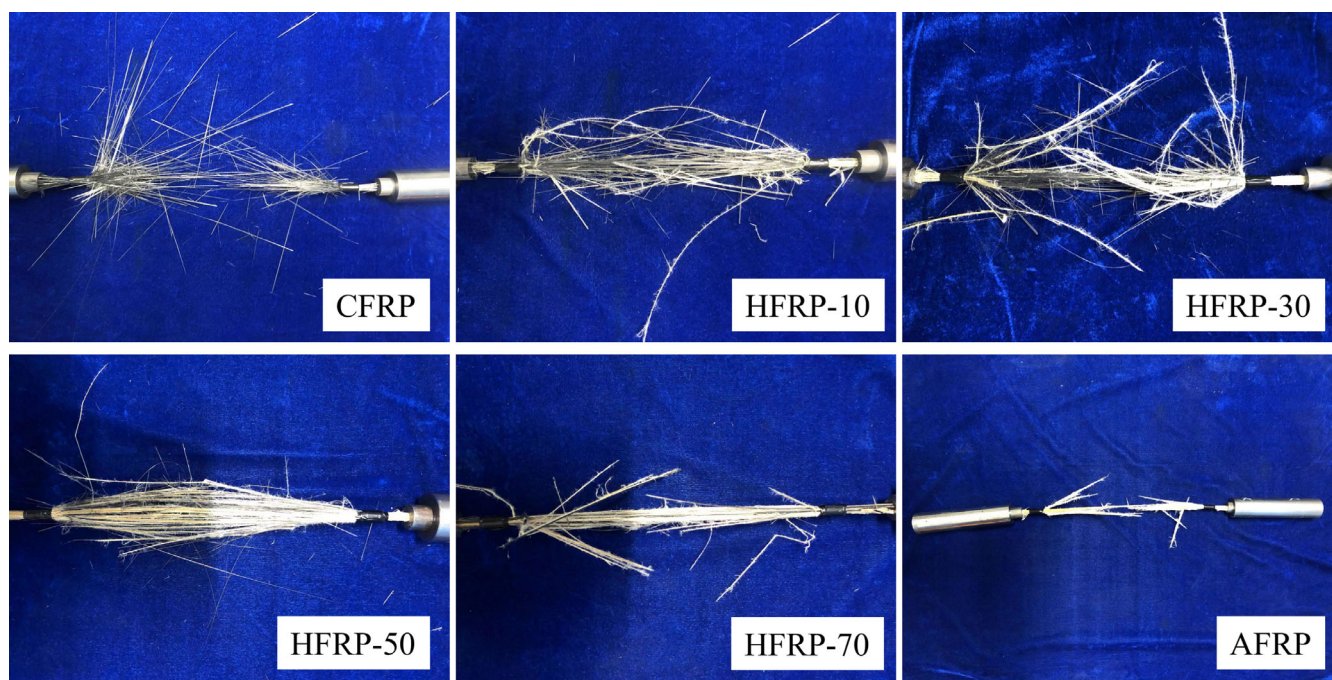
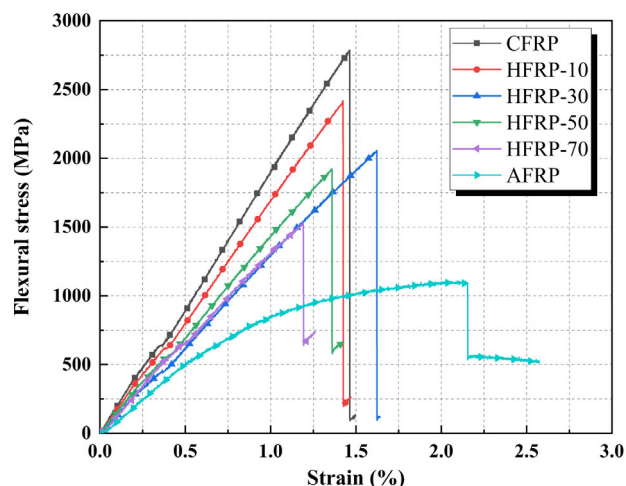


FIGURE 3 Tensile failure photos of fiber-reinforced polymer composites.

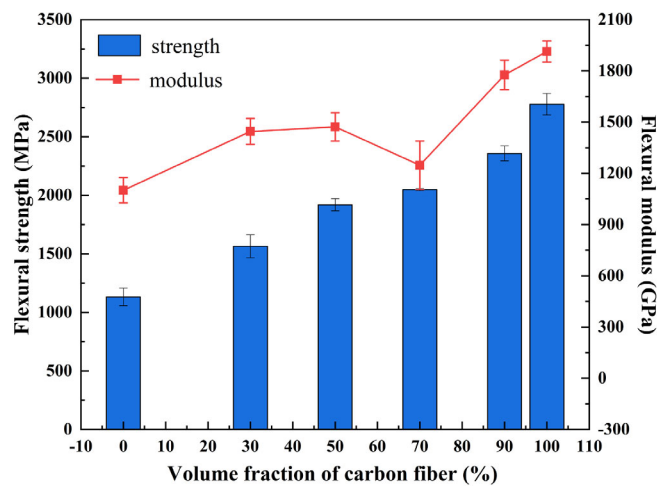
increase in carbon fiber content, the flexural strength and modulus of A/CHFRP showed a steady improvement. The flexural properties of A/CHFRP are primarily affected by the compression performance of the surface carbon fibers. Consequently, an increase in carbon fiber content leads to higher strength and modulus in A/CHFRP.

Figure 5 shows the failure photographs of the CFRP, AFRP, and A/CHFRP under flexural loads. Obvious indentations appeared on the upper surfaces of CFRP,

AFRP, and HFRP, showing a compression failure mode. Apparent fiber breakage and interface debonding appeared on the surface of the CFRP, and the fracture was relatively neat, showing evident brittle fracture characteristics. A white crack appeared on the surface of the AFRP, and debonding cracks appeared at the interface; however, no fiber breakage occurred. HFRP-10, HFRP-50, and HFRP-70 still exhibited debonding. In addition, as the carbon fiber content increased, the cracks of carbon fiber breakage gradually became less noticeable,



(A) stress-strain



(B) strength and modulus

FIGURE 4 (A) Flexural stress-strain and (B) flexural strength and modulus of fiber-reinforced polymer composites.

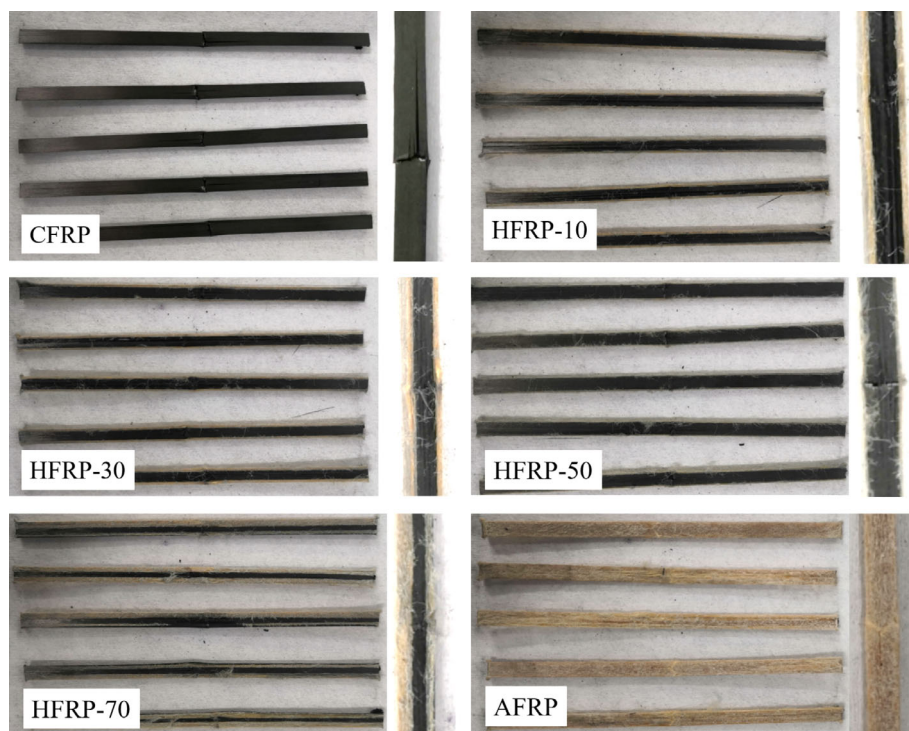


FIGURE 5 Flexural failure photos of fiber-reinforced polymer composites.

whereas the cracks in the aramid fiber layer due to deformation gradually became noticeable.

3.3 | Shear properties

As shown in Figure 6, the shear strength of the A/CHFRP gradually decreased with increasing carbon fiber content. The shear strengths of HFRP-10, HFRP-30, HFRP-50, and HFRP-70 were 1.145%, 4.18%, 4.73%, and 10.09% higher, respectively, than those of the CFRP. Aramid fibers exhibited better shear performance than carbon fibers. The higher the aramid content, the less

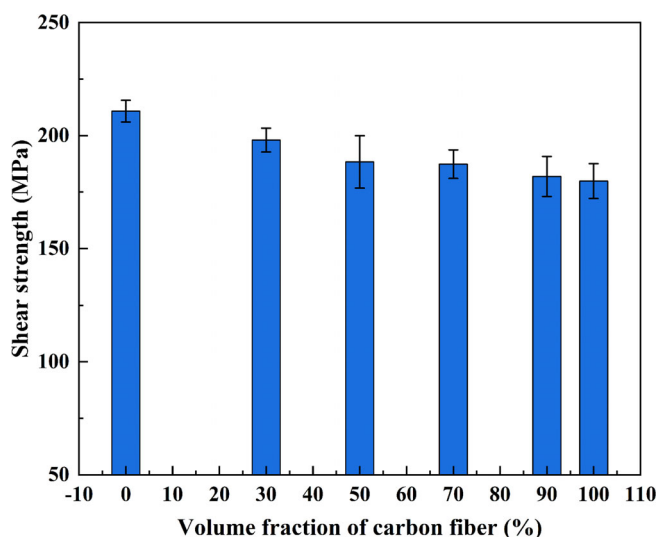


FIGURE 6 Shear strength of fiber-reinforced polymer composites.

likely HFRP is to be damaged during a shear test. Therefore, as the aramid fiber content increased and the carbon fiber content decreased, the shear strength of the hybrid composites increased. Para-aramid fibers are microscopic skin-core structures. The core layer is often highly ordered and crystalline, providing strength and stiffness to the aramid fiber, whereas the skin layer is a non-crystalline area, providing toughness to the aramid fiber. A heating method and damage to the skin area during the pultrusion process cause the shear resistance of the aramid fiber to decrease. Therefore, the shear performance of the A/CHFRP was not significantly improved compared to that of the CFRP.

Figure 7 shows the failure photographs of A/CHFRP under shear stress. The shear failure section of the CFRP was relatively neat, indicating brittle failure. The shear failure fracture of the AFRP was white, and the aramid fibers were not completely disconnected. This was mainly attributed to the superior toughness and shear resistance of aramid fibers. With the gradual increase in aramid fiber content, the shear failure sections of HFRP-10, HFRP-30, HFRP-50, and HFRP-70 progressively approached that of AFRP, resulting in a corresponding increase in strength.

Before applying hybrid composite materials to structures, conduct comprehensive material property tests, including assessments of thermal stability, chemical resistance, and UV stability, to ensure that the selected material combination can meet the demands of specific environmental conditions. A mechanical property prediction model incorporating the fiber volume ratio is developed through theoretical derivation. Experimental tests, such as tension, compression, and bending tests, are

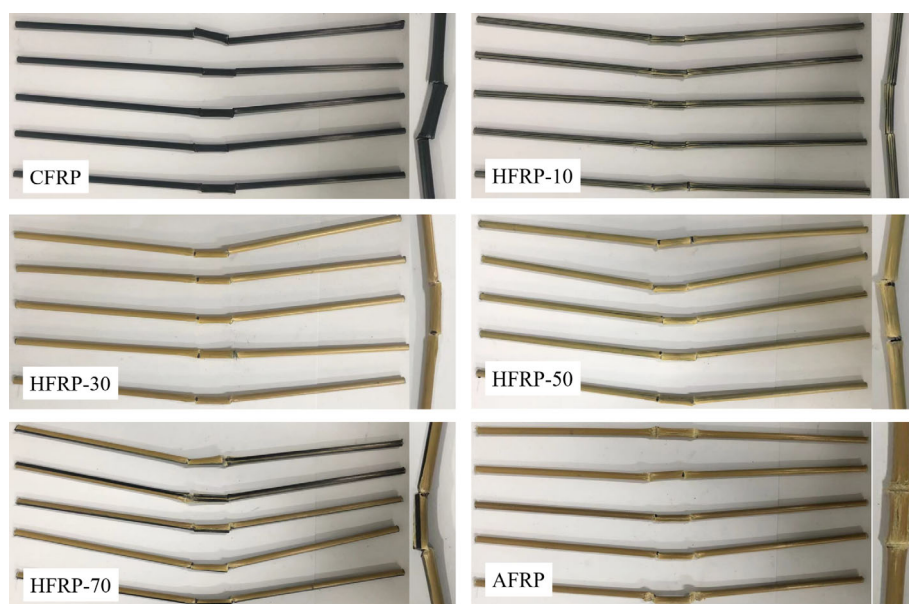


FIGURE 7 Shear failure photos of fiber-reinforced polymer composites.

conducted to evaluate the mechanical properties of composite materials with varying fiber volume ratios. These tests provide crucial data to validate simulation results and optimize fiber ratios. Additionally, fatigue and durability tests are performed, particularly for structural applications like bridge cables that are subjected to cyclic loads. Fatigue testing is essential in evaluating the long-term performance of materials under repeated loading and for further refining the fiber volume ratio.

4 | CONCLUSIONS

This study fabricated A/CHFRP bars through the pultrusion process, utilizing aramid fibers as the outer layer and carbon fibers as the inner core. The static mechanical properties of the CFRP and AFRP bars were compared, and the influence of the aramid fiber to carbon fiber volume ratio on the tensile, flexural, and shear properties of the A/CHFRP bars was analyzed to fully understand the performance of the C/AFRP bars.

The tensile strength of A/CHFRP satisfied the triangular prediction area surrounded by the mixed and bilinear mixed laws. However, the tensile stress-strain curve of A/CHFRP did not display pseudo-ductility. Under bending loads, A/CHFRP primarily experienced carbon fiber layer crushing and interlayer debonding, resulting in a flexural limit strain lower than that of CFRP. As the volume of carbon fiber in A/CHFRP increased, the flexural strength and modulus of A/CHFRP improved. Heating reduced the toughness of aramid fibers and deteriorated their shear resistance. The shear resistance of A/CHFRP increased with increasing aramid content, but the improvement was not significant compared with that of CFRP.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

- Feng B, Wang X, Wu Z, Yang Y, Pan Z. Performance of anchorage assemblies for CFRP cables under fatigue loads. *Structure*. 2021;29:947-953. doi:10.1016/j.istruc.2020.12.012
- Qi X, Tian J, Guo R, Xian G. Hydrothermal aging of carbon fiber reinforced polymer rods intended for cable applications in civil engineering. *J Mater Res Technol*. 2023;26:5151-5166. doi:10.1016/j.jmrt.2023.08.285
- Wang Y, Zheng YQ, Zhang WH, Lu QR. Analysis on damage evolution and corrosion fatigue performance of high-strength steel wire for bridge cable: experiments and numerical simulation. *Theor Appl Fract Mec*. 2020;107:102571. doi:10.1016/j.tafmec.2020.102571
- Fan Z, Ye Q, Xu X, Ren Y, Huang Q, Li W. Fatigue reliability-based replacement strategy for bridge stay cables: a case study in China. *Structure*. 2022;39:1176-1188. doi:10.1016/j.istruc.2022.03.093
- Yang Y, Fahmy MFM, Guan S, Pan Z, Zhan Y, Zhao T. Properties and applications of FRP cable on long-span cable-supported bridges: a review. *Compos Part B-Eng*. 2020;190:107934. doi:10.1016/j.compositesb.2020.107934
- Chen W, Yang S, Zhang X, Jordan ND, Huang J. Embodied energy and carbon emissions of building materials in China. *Build Environ*. 2022;207:108434. doi:10.1016/j.buildenv.2021.108434
- Burchart-Korol D. Life cycle assessment of steel production in Poland: a case study. *J Clean Prod*. 2013;54:235-243. doi:10.1016/j.jclepro.2013.04.031
- Meier U. Carbon fiber reinforced polymer cables: why? Why not? What if? *Arab J Sci Eng*. 2012;37:399-411. doi:10.1007/s13369-012-0185-6
- Guo R, Li C, Niu Y, Xian G. The fatigue performances of carbon fiber reinforced polymer composites—a review. *J Mater Res Technol*. 2022;21:4773-4789. doi:10.1016/j.jmrt.2022.11.053
- Li H, Xu S, Zhang Z, Song C. Experimental and numerical investigation on the corrosion effects on the bonding behavior between CFRP and steel. *Compos Struct*. 2021;259:113465. doi:10.1016/j.compstruct.2020.113465
- Wang L, Zhang J, Xu J, Han Q. Anchorage systems of CFRP cables in cable structures—a review. *Construct Build Mater*. 2018;160:82-99. doi:10.1016/j.conbuildmat.2017.10.134
- Xin M, Zhang Y, Guo Z, et al. Design of novel glass fiber reinforced polypropylene cable-anchor component and its long-term properties exposed in alkaline solution. *Case Stud Constr Mat*. 2024;20:e03383. doi:10.1016/j.cscm.2024.e03383
- Wang A, Xu G, Liu X. Effect of polyurea coating on low-velocity impact properties of unidirectional carbon fiber-reinforced polymer composites plates. *Structure*. 2024;61:106090. doi:10.1016/j.istruc.2024.106090
- Wang Z, Xian G. Impact performances of fiber reinforced polymer composites and cables: a review. *Compos Struct*. 2023;319:117128. doi:10.1016/j.compstruct.2023.117128
- Wu Q, Zhi X, Li Q, Guo M. Experimental and numerical studies of GFRP-reinforced steel tube under low-velocity transverse impact. *Int J Impact Eng*. 2019;127:135-153. doi:10.1016/j.ijimpeng.2019.01.010
- Zhang W, Jia L, Luo K, Teng X. A novel anchor method for large tonnage CFRP cable: Anchorage design and full-scale experiment. *Compos Struct*. 2024;336:118004. doi:10.1016/j.compstruct.2024.118004
- Wang Z, Guo R, Zhang P, et al. Transverse low-velocity impact behaviors of pultruded carbon-fiber-reinforced polymer rods with tensile preloads: experiment and simulation. *Compos Part B-Eng*. 2024;283:111672. doi:10.1016/j.compositesb.2024.111672

18. Fang Y, Fang Z, Liu P, Xiang Y, Zhou X. Experimental investigation on post-impact bond performance of carbon fiber-reinforced polymer tendons in ultra-high performance concrete. *Construct Build Mater*. 2024;427:136214. doi:10.1016/j.conbuildmat.2024.136214
19. Guo R, Xian G, Li F, Li C, Hong B. Hygrothermal resistance of pultruded carbon, glass and carbon/glass hybrid fiber reinforced epoxy composites. *Construct Build Mater*. 2022;315:125710. doi:10.1016/j.conbuildmat.2021.125710
20. Muthalagu R, Kumar SS, Pati PR, et al. Influence of Prosopis juliflora bark powder/fillers on the mechanical, thermal and damping properties of jute fabric hybrid composites. *J Mater Res Technol*. 2024;33:3452-3461. doi:10.1016/j.jmrt.2024.10.066
21. Goyal A, Melenka GW. Investigation of carbon-aramid hybrid braided composites using digital volume correlation. *Compos Struct*. 2023;321:117207. doi:10.1016/j.compstruct.2023.117207
22. Mostafa NH, Ismarrubie ZN, Sapuan SM, Sultan MTH. Effect of fabric biaxial prestress on the fatigue of woven E-glass/polyester composites. *Mater Design*. 2016;92:579-589. doi:10.1016/j.matdes.2015.12.109
23. Mostafa NH, Ismarrubie ZN, Sapuan SM, Sultan MTH. Effect of equi-biaxially fabric prestressing on the tensile performance of woven E-glass/polyester reinforced composites. *J Reinf Plast Comp*. 2016;35(14):1093-1103. doi:10.1177/0731684416638553
24. Wu L, Wang W, Jiang Q, Lin JH, Tang Y. Illustrating hybrid effect and damage evolution of carbon/aramid braided composite under low-velocity impact. *Compos Struct*. 2020;245:112372. doi:10.1016/j.compstruct.2020.112372
25. Guo R, Xian G, Li C, Hong B. Effect of fiber hybrid mode on the tension-tension fatigue performance for the pultruded carbon/glass fiber reinforced polymer composite rod. *Eng Fract Mech*. 2022;260:108208. doi:10.1016/j.engfracmech.2021.108208
26. Djerrad A, Fan F, Zhi X, Wu Q. Experimental and numerical studies on AFRP-reinforced thin-walled tubes under axial impact loading. *Int J Impact Eng*. 2024;190:104974. doi:10.1016/j.ijimpeng.2024.104974
27. Xu J, Tang L, Liu Y, et al. Hybridization effects on ballistic impact behavior of carbon/aramid fiber reinforced hybrid composite. *Int J Impact Eng*. 2023;181:104750. doi:10.1016/j.ijimpeng.2023.104750
28. Wang H, Lin Y, Jiang H, Liu Z. Inter-layer failure and toughening mechanisms of carbon/aramid hybrid fiber composites interleaved with micro/nano pulps under low-velocity impact load. *Thin Wall Struct*. 2024;202:112086. doi:10.1016/j.tws.2024.112086
29. Wagih A, Sebaey TA, Yudhanto A, Lubineau G. Post-impact flexural behavior of carbon-aramid/epoxy hybrid composites. *Compos Struct*. 2020;239:112022. doi:10.1016/j.compstruct.2020.112022
30. Aredla R, Dasari HC, Kumar SS, Pati PR. Mechanical properties of natural fiber reinforced natural and ceramic fillers for various engineering applications. *Interactions*. 2024;245(1):249. doi:10.1007/s10751-024-02109-3
31. Purohit A, Dehury J, Sitani A, et al. A novel study on the stacking sequence and mechanical properties of jute-Kevlar-epoxy composites. *Interactions*. 2024;245(1):100. doi:10.1007/s10751-024-01946-6
32. Swolfs Y, Gorbatiikh L, Verpoest I. Fibre hybridisation in polymer composites: a review. *Compos Part A-Appl S*. 2014;67:181-200. doi:10.1016/j.compositesa.2014.08.027
33. Zhang C, Rao Y, Li W. Low-velocity impact behavior of intra-layer hybrid composites based on carbon and glass non-crimp fabric. *Compos Struct*. 2020;234:111713. doi:10.1016/j.compstruct.2019.111713
34. Kumar SS, Shyamala P, Pati PR, Giri J, Makki E, Sathish T. Mechanical (static and dynamic) characterization and thermal stability of hybrid green composites for engineering applications. *J Mater Res Technol*. 2024;30:7214-7227. doi:10.1016/j.jmrt.2024.05.132
35. Dash S, Satpathy MP, Routara BC, Pati PR, Gantayat S. Enhancing mechanical and tribological performance of hybrid composites: an experimental study utilizing response surface methodology and firefly algorithm. *Polym Compos*. 2024;45:15924-15940. doi:10.1002/pc.28880
36. Zachariah SA, Shenoy BS, Pai KD. Comprehensive analysis of in-plane tensile characteristics of thin carbon/aramid hybrid composites using experimental and RVE-based numerical study. *Compos Struct*. 2021;271:114160. doi:10.1016/j.compstruct.2021.114160
37. Wang A, Yang Y, Zhang P, Liu X, Xu G. Effect of fiber volume fraction on hybrid effects of tensile properties of unidirectional aramid/carbon fiber hybrid composites. *Polym Compos*. 2024;45(3):2050-2062. doi:10.1002/pc.27903
38. Song JH. Pairing effect and tensile properties of laminated high-performance hybrid composites prepared using carbon/glass and carbon/aramid fibers. *Compos Part B-Eng*. 2015;79:61-66. doi:10.1016/j.compositesb.2015.04.015
39. Bao J, Wang Y, An R, Cheng HW, Wang FC. Investigation of the mechanical and ballistic properties of hybrid carbon/aramid woven laminates. *Def Technol*. 2022;18(10):1822-1833. doi:10.1016/j.dt.2021.09.009
40. Mackin CD. *Pultrusion Manufacturing, Materials Testing, and Life Cycle Assessment of Thermoplastic GFRP Composites*. Master thesis of University of New Brunswick; 2021.
41. Wang A, Liu X, Yue Q, Xian G. Effect of volume ratio and hybrid mode on low-velocity impact properties of unidirectional flax/carbon fiber hybrid reinforced polymer composites. *Thin Wall Struct*. 2023;187:110764. doi:10.1016/j.tws.2023.110764
42. Huang B, Ma M, Liu X, et al. Investigation on the fundamental mechanical properties and probabilistic characteristics of unidirectional carbon fiber reinforced polymer composite plates [J]. *Polym Test*. 2024;131:108355. doi:10.1016/j.polymertesting.2024.108355
43. Bai N, Li H, Lan C, Spencer BF Jr. Influencing factors and sensitivity analysis for the fatigue of FRP wire based on the progressive fatigue damage model. *Compos Struct*. 2024;334:117982. doi:10.1016/j.compstruct.2024.117982
44. Huang B, Yang Y, Liu X, Yue QR. Study on probability model and length effect of tensile strength of unidirectional CFRP plate. *Compos Struct*. 2023;324:117531. doi:10.1016/j.compstruct.2023.117531
45. Fuller JD, Wisnom MR. Pseudo-ductility and damage suppression in thin ply CFRP angle-ply laminates. *Compos Part A-Appl S*. 2015;69:64-71. doi:10.1016/j.compositesa.2014.11.004
46. Yu HN, Longana ML, Jalalvand M, Wisnom MR, Potter KD. Pseudo-ductility in intermingled carbon/glass hybrid composites

- with highly aligned discontinuous fibres. *Compos Part A-Appl S*. 2015;73:35-44. doi:[10.1016/j.compositesa.2015.02.014](https://doi.org/10.1016/j.compositesa.2015.02.014)
47. Czél G, Pimenta S, Wisnom MR, Robinson P. Demonstration of pseudo-ductility in unidirectional discontinuous carbon fibre/epoxy prepreg composites. *Compos Sci Technol*. 2015;106: 110-119. doi:[10.1016/j.compscitech.2014.10.022](https://doi.org/10.1016/j.compscitech.2014.10.022)
48. Wang A, Liu X, Yue Q, Xian G. Tensile properties hybrid effect of unidirectional flax/carbon fiber hybrid reinforced polymer composites. *J Mater Res Technol*. 2023;24:1373-1389. doi:[10.1016/j.jmrt.2023.03.078](https://doi.org/10.1016/j.jmrt.2023.03.078)

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